



Edward Tufte's Visual Encoding Transcript

Hello everyone, welcome to BITS Pilani Digital. Today, we will continue our learning journey for the course Data Visualisation and Storytelling. I will be taking you to a new module, Designing Visuals and the Best Practises therein.

Today, we will be learning Edward Tufte's Visual Encoding Practises and Conversion of Data into Visualisations to Draw Valuable Insights. In this lesson, we shall be learning about the basic principles on how to design visualisations and what care should be taken so that you can effectively communicate with your audience. Before we dwell into the content, let me take a pause and explain the different types of data types that you would encounter as a data visualisation task.

Before we start on our journey, let us understand the different types of data. In a data visualisation exercise, you often come across different types of numerical data. They are categorised into four types.

One is the variables that represent quantitative variables like gender or for example race. These are specific variables which might take values 1, 2, 3, 4, but per se the numbers don't have any meaning. They represent the category.

For example, you can code male as 1 and female as 0 or vice versa. 1 and 0 per se do not have any meaning. They just represent it belongs to a particular category.

Such variables are called nominal variables. There are other kinds of variables like ordinal variables. When you ask a customer to give review on your product, he would rate it on a scale of 1 to 5. Such rankings say 1 is lower, 5 being higher would reflect some amount of preference towards the product.

Such variables are called ordinal variables which inherently carry some order in them. Therefore, the name ordinal. However, like in case of nominal variables, ordinal variables also represent likeliness or some amount of ordering.

Again, you can only say some product has higher ratings than the other product or some customer has rated your product. For example, customer A has reviewed your product by giving 5 stars while customer B has given a 3 star review. All that you can say is that customer A prefers your product more than customer B, but you cannot say he likes it by a factor of 2. You cannot make any arithmetical calculations here and say this is so much order higher.

Just you can say that it is larger or smaller in sense of preference or likeliness or whatever ordering you are trying to talk about. Such variables are called ordinal variables. All that you can do a comparison is whether it is liked similarly, whether it is liked highly, whether it is liked lowly, that is it, whether it is liked less, whether it is liked less.

So, these concepts are useful in making comparisons, nominal variable, then ordinal



variable. Then we have ratio interval scale. Then we have interval scale.

Interval scales are defined over an interval range. Common example is the temperature measurement in degrees centigrade or in Fahrenheit. For example, when you have say the temperature is 50 degrees in Hyderabad and 25 degrees in Shimla, does not mean Hyderabad is twice as hot as Shimla.

All that we can say is the temperature difference between Hyderabad and Shimla is 25 degrees centigrade. That is it. You can make comparisons like how much is the temperature and also you can see the distance between two categories.

For example, Shimla is having 25 degrees centigrade temperature, Hyderabad is having 25 degrees more temperature, but you cannot make a statement like it is twice. You cannot do proportional analysis with the interval scale. Finally, you have ratio type data variables where you can perform all types of numerical calculations like addition, subtraction, and proportion, and you can do all the comparisons.

These are the variables we are generally interested in types of revenue, profit, growth, and then quantitative variables you generally fit into these values. All other variables are also encountered, but they have specific purposes or they capture specific characteristics of data. So whenever we are dealing with a data visualisation exercise, we should be aware of the underlying data variable type so that we will not apply the wrong arithmetic and draw spurious conclusions.

It has to be, care has to be exercised while we are exploring the data and identify the data type and use the appropriate arithmetic function that is allowed for such variable type. Now, let us get into another tough case, visual encoding principles. Whenever we see a graph or a visualisation, our mind is automatically performing three tasks.

This is how humans decode the information that is there in the visual. Let us say we are talking about a graph. There are three tasks as I was mentioning detection, assembly, and estimation.

So whenever we are seeing a graph, first thing that we try to do is the detect the geometric object or the one that reflects the quantity in the graph. So we are trying to look out for the physical resemblance of some quantity in the graph either through for example, in a bar graph to the length of the bar or if you are talking about a pie chart, it will be the area of the pie that you are interested for a particular segment so on and so forth. This is called detection.

Detection pertains to the ability of brain to assign meaning to the physical objects that are there in the graph. Next comes assembly. Assembly is when the human mind looks for the continuity or the grouping of the geometric objects in the graph.

Assembly is when the brain is trying to decode the geometric objects. We now move



to the next level of perception which the humans use to decode the information that is in the visuals or graphs. This is called assembly.

Whenever we look at the graph, first we try to detect the geometrical objects. The next step that we try to see, we try to understand how these geometrical objects are grouped or aligned or assembled on the visual space. So the assembly follows closely the just like principle of continuity.

The continuity principle says that when a human mind sees a continuous series, it automatically decides what is the underlying relationship or imposes an underlying relationship. But this is the reason when you have a graphic and it is ordered in a logical fashion, you will try to convince the audience much more faster that there is a trend that you are trying to reflect. In this case, we are seeing the voter margin across constituencies and we can see it is arranged in the descending order by the margin of votes that the winning candidate has secured.

For example, in this case, Faridabad has the highest voter margin as compared to Gurgaon or Hisar which has the least margin. So these kind of conclusions can be immediately drawn because it is in a descending voter logical order. Inherently, what are you doing while doing the detection and assembly, you are also doing the estimation.

You are trying to compare the quantities that are being told by these objects in particular order. So this is estimation. When we are trying to compare or give some sense to the comparison like how large or how different are two particular categories are on the visual.

Whenever we are decoding a graph or a visual, these three functions are being performed. Now this begs the question when you are designing the visual, you have to keep in the mind the sequence and the efficiency or the ease with which these three functions can be performed by your audience. You should help your audience to do these three functions detection, assembly and estimation in an efficient manner.

Now we will see how to ensure that you enable these functions are done by your audience in an efficient manner. So first of all, you have to decide what is the comparison or what is the information that you want to highlight in the graphic. And then you have to decide how will you portray it or position it on the graphic so that the detection, assembly and estimation functions are done very properly.

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